

Pascal Jutras-Dubé

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Education

PhD Computer Science	Purdue University	2023 – expected May 2028
MSc Computer Science	Université de Montréal	2020 – 2022
BSc Mathematics and Computer Science	Université de Montréal	2017 – 2020

Fin-ML NSERC CREATE Scholarship — national training program in machine learning for quantitative finance (2020).

Research Experience

Research Assistant, Purdue University Jan 2023 – Present

- **Sticky Jump Diffusions** [[arXiv](#)] [[code](#)] (under review, NeurIPS 2026): derived the exact time reversal of a continuous-time jump–diffusion by flux balance, and proved its intractable reverse jump intensity equals a conditional expectation of forward-known quantities, estimated by a Rao–Blackwellized reweighting of one cross-entropy classifier with no auxiliary network. Training is simulation-free via inverse-CDF sampling of jump times and closed-form Gaussian convolutions. Cuts CIFAR-10 FID 15.9 → 14.6 and lifts Sudoku exact-match accuracy 47% → 96% at 4× faster convergence over the strongest prior hybrid under matched compute.
- **One-step diffusion samplers** [[arXiv](#)] [[code](#)] (AISTATS 2026): learned a step-conditioned ODE so one large step reproduces a many-step trajectory, and derived a deterministic-flow importance weight that keeps evidence (ELBO) estimates stable where standard integrators break down. Matches multi-step sample quality and estimator accuracy at 1/256 of the network evaluations.

Research Intern, National Bank of Canada Jun 2021 – Nov 2022

- Built a quantitative scoring framework (record linkability, information gain) measuring the privacy–utility frontier of customer data-release strategies; published at HICSS 2024 and in the *Journal of Business Analytics* (2025).

Teaching Assistant, Purdue University Jan 2023 – Present

- Graduate Statistical Machine Learning: wrote and graded assignments and exams, and delivered guest lectures on diffusion models.

Software Engineer Intern, GRAITEC May 2019 – Aug 2019

- Shipped full-stack features for a production woodworking CAD application in an agile team.

Selected Publications

Jutras-Dubé, P., Pynadath, P., Lu, J., Gao, Y., & Zhang, R.

Sticky Jump Diffusions: A Unifying View of Masked, Continuous, and Hybrid Diffusion. Under review at NeurIPS, 2026. [[code](#)]

Jutras-Dubé, P., Zhang, J., Wang, Z., & Zhang, R.

One-Step Diffusion Samplers via Self-Distillation and Deterministic Flow. AISTATS, 2026. [[code](#)]

Lee, S., Jutras-Dubé, P., Wen, B., & Zhang, R.

Neural Discrete Controlled Monte Carlo Samplers. ProbML Workshop @ ICML, 2026

Jutras-Dubé, P., Al-Khasawneh, M. B., Yang, Z., Bas, J., Bastin, F., & Cirillo, C.

Copula-Based Transferable Models for Synthetic Population Generation. Transportation Research Part C, 2024

Full list (incl. IROS 2024–2025, ICLR workshop 2025, HICSS 2024, J. Business Analytics 2025): [Google Scholar](#).

Skills

Quantitative: SDEs and jump processes, time reversal of Markov processes, point-process and hazard-rate modeling, Monte Carlo variance reduction (Rao–Blackwellization, importance sampling), inverse-CDF simulation, copulas, Bayesian inference, stochastic optimization

Engineering: Python, JAX (multi-GPU), PyTorch, NumPy/pandas, Hydra, Weights & Biases, Slurm, Git

Additional

Awards: Fin-ML NSERC CREATE Scholarship (2020); UdeM Graduate Scholarship (2021); DIRO Excellence (4×, 2020–2022)

Reviewer: NeurIPS 2026, ICML 2026 (Silver Reviewer Award), ICLR 2025–2026, AISTATS 2026, AAAI 2027

Languages: French (native), English (fluent)

Almost Surely: Founder of a fashion label for the STEM community, almost-surely.com